The topology of the tested system is illustrated below:

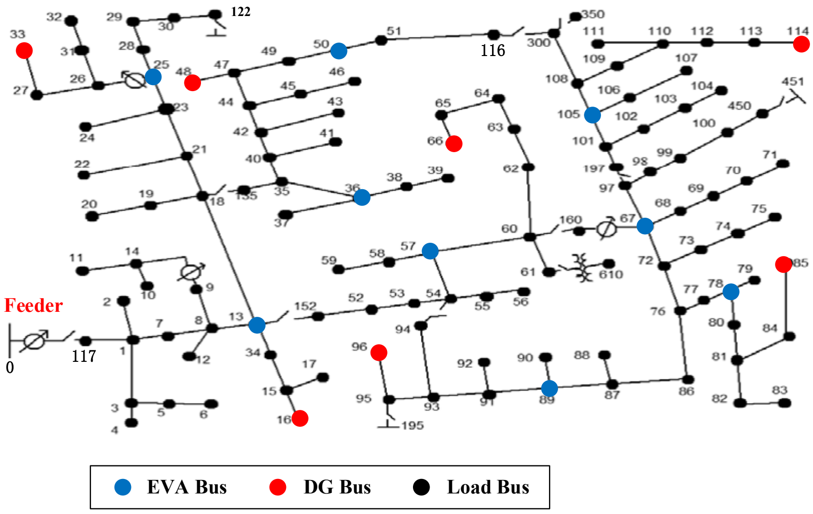

Fig. A1. The modified IEEE 123-bus distribution system.

To further examine the scalability of the proposed framework, a case study on IEEE 123-bus system is conducted and the results are summarized in Table A.I.

TABLE A.I

PERFORMANCE OF DSO-EVA COORDINATION IN IEEE 123-BUS SYSTEM UNDER DIFFERENT METHODS

	TC ($)	EC ($)	RC ($)	EVC ($)	VR	CT (s)
M1	7995.0	5003.4	1246.1	1745.5	7.90%	1024.25
M2	8026.7	5003.4	1277.8	1745.5	7.60%	7200
M3	8061.9	5003.4	1313.0	1745.5	7.00%	1.52
M4	8099.6	5003.4	1350.7	1745.5	7.00%	3.68
M5						Infeasible
M6						Infeasible

* TC: Total Cost; EC: Energy Cost; RC: Reserve Cost; EVC: EVA Cost; VR: Violation Rate; CT: Computational Time.

M1 achieves the lowest total operating cost of $7995.0 among all feasible methods, outperforming the mean-value box method (M3) and the ellipsoid-reconstructed box method (M4) by $66.9 and $104.6, respectively. Since energy and EVA costs are identical across M1–M4, the entire cost saving derives from a reduction in reserve cost: M1 lowers the reserve cost by 5.1% relative to M3 and by 7.7% relative to M4. This confirms that the proposed OUSL framework can improve DSO–EVA dispatch economy by learning less conservative and operation-oriented load uncertainty envelopes.

M2 also uses the OUSL formulation but computes the gradient through KKT-based sensitivity analysis. Its total cost is lower than those of M3 and M4, but higher than that of M1. More importantly, its computational time reaches the limit of 7200 s, which is about seven times that of M1. This indicates that explicit KKT-based sensitivity construction is not suitable for large-scale systems, whereas the proposed dual-sensitivity gradient provides a more scalable implementation of the same problem-oriented learning principle.

The generic ellipsoid method M5 and the DRJCC method M6 become infeasible in the IEEE 123-bus system. This is mainly because these methods construct uncertainty sets from generic geometric or imprecise moment-based information. As the system dimension increases, such uncertainty sets can become overly conservative and require reserve capacities beyond the available network resources.

Fig. A2 illustrates the objective value of M1 during the OUSL learning process. The objective decreases monotonically from approximately 8063 to 7995. The sharp reduction in the first few iterations indicates that the initial mean-value box contains significant unnecessary conservatism. The subsequent stepwise decrease reflects the non-smooth nature of the DSO dispatch problem, where the active reserve and network constraints change as the load-envelope shape is updated. Overall, the results demonstrate that the dual-sensitivity-based OUSL procedure can steadily improve the dispatch objective in a high-dimensional distribution-network setting.

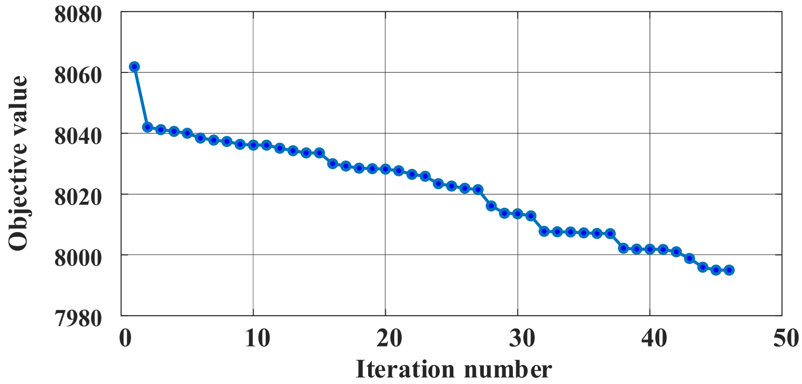

Fig. A2. Objective-value iteration trajectory of the proposed OUSL method on the IEEE 123-bus system.

It should be noted that the computational time of M1 includes the offline OUSL training process. This offline cost should be distinguished from the online scheduling cost. Once the optimized OUSL envelope is learned from historical samples, the online DSO-EVA dispatch only needs to use the fixed learned envelopes, and its deterministic reformulation has the same structure as the mean-value and ellipsoid-reconstructed box methods. Therefore, the proposed framework improves dispatch economy without increasing the structural complexity of the online scheduling problem.